Proposed Solution

Multimodal Attendance & Engagement Agent • Nirbhay Dubey | B.Tech CSE (Year 1)

1. Solution Summary

A privacy-preserving, local AI application that tracks classroom attendance through face detection and recognition, while calculating student attentiveness from 3D head pose, eye gaze, and classroom acoustics, then delivers real-time teaching recommendations. It runs on a standard laptop connected to a classroom camera and sends no data to the cloud.

2. Key Components

Component	What it does
Vision pipeline	Haar cascade detection, DeepSort tracking, Facenet recognition, MediaPipe head pose and iris gaze
Audio pipeline	Librosa features and an adaptive-noise-floor classifier (QUIET, SPEECH, MODERATE_NOISE, DISTURBANCE)
Fusion and scoring	Temporal sync buffer aligns video and audio; engagement = 0.40 gaze + 0.20 pose + 0.20 presence + 0.20 audio, smoothed by EMA (beta 0.70)
Attendance service	Accumulates presence seconds; PRESENT at 75% or more, PARTIAL at 40-74%, ABSENT below 40%
Recommendation engine	Five rules: low engagement, acoustic disturbance, gaze aversion, optimal attention window, low attendance; 30 s debounce
Backend and storage	FastAPI with REST, MJPEG, and 4 Hz WebSocket telemetry; SQLite via SQLAlchemy; JSON index and CSV exports
Dashboard	Streamlit app with 8 pages from live monitoring to settings

3. How It Works (Lecture Flow)

- The lecturer starts a session. The camera and (optionally) audio chunks stream into the backend.
- Faces are detected and tracked; enrolled students are recognized, others stay as anonymous IDs.
- Pose, gaze, and audio state are fused each frame to produce engagement scores.
- Presence time accumulates per student; recommendations are generated when rules trigger.
- On "Finalize & End Session", verdicts are saved to SQLite and a CSV can be exported.

4. Unique Selling Points

Feature	Benefit
Offline edge processing	No cloud transmission, so privacy by design and no recurring fees
Presence-over-time attendance	Harder to cheat than one-off roll calls
Consent-first biometrics	Non-consenting people are never enrolled; deregistration purges data
Multimodal engagement index	Combines vision and sound instead of relying on one signal
Explainable rule engine	Every suggestion has a named trigger and action steps
Human override	Lecturer stays in control with an audit reason

5. Expected Impact

- Recovers the 5-10 minutes per lecture otherwise spent on roll call.
- Reduces proxy attendance by requiring sustained visual presence.
- Gives lecturers objective, live feedback on class attention.
- Shows that edge AI can meet privacy rules such as GDPR, FERPA, and India's DPDP Act 2023 by keeping data local.

6. Feasibility and Scalability

The system is feasible because it uses open-source libraries on ordinary hardware, and its 12 automated test suites pass. Scaling to large halls would use multi-camera RTSP streams, a time-series database for per-frame telemetry, and optional on-device language models for lecture summaries.